An amateur, pre-registered monitoring study of asteroid 99942 Apophis: impact exclusion, agency agreement and model predictions for the 13 April 2029 close approach

Project Hannibal · data 2026-10-23T21:03:00Z to 2026-10-23T21:59:05Z · 2 verified snapshots · report built 2026-10-23 22:08 UTC from commit f1a5820

Abstract

We monitored the public orbit products of NASA/JPL and ESA NEOCC for asteroid 99942 Apophis, hourly, from 2026-10-23T21:03:00Z to 2026-10-23T21:59:05Z (2 verified snapshots, every raw reply hashed with SHA-256). Before collecting data we fixed eight criteria (C1 to C8) for what would count as real reason for concern: risk listings, a pass reaching the atmosphere, disagreement between the agencies, an impact solution at a point off Lisbon, a shift of the 2029 distance, and a push the Yarkovsky model doesn't explain. C1 to C6 are evaluated automatically. Over the period, JPL's predicted 2029 distance ranged 38,011.5 to 38,011.5 km and ESA's 38,013.8 to 38,013.8 km. Orbit solutions seen: 220. No pre-registered criterion fired in any verified snapshot. Over the study period, the public data from NASA/JPL and ESA NEOCC give no evidence for a risk of impact by 99942 Apophis, including at the point off Lisbon, and no evidence for a push the Yarkovsky model doesn't explain. This is a null result for the period covered. It does not test predictions that depend on observations not yet made (Apophis is next observable from about 20 December 2026).

1 Question and pre-registration

Question: is there evidence, in public data from independent professional sources, that Apophis poses a real risk of impact, including at 38.70N 9.60W off Lisbon during the 2029 flyby? The criteria, two hypotheses and a recorded bet were committed to git before the data that tests them existed (`docs/investigation/apophis/PREREGISTRATION.md`; registration commit 36f3703, 23 Oct 2026, 21:03 UTC).

Criterion	Definition (abridged)	Checked
C1	JPL Sentry or ESA's risk list carries Apophis	automatic
C2	Any pass's minimum possible distance ≤ Earth radius + 100 km (JPL or ESA)	automatic
C3	JPL and ESA disagree by > 3 combined sigma on a pass whose window is smaller than its distance	automatic
C4	Impact at the point off Lisbon is POSSIBLE on JPL's 3-sigma bounds	automatic (hourly collector)
C5	A new JPL solution moves the 2029 distance by > 3.3 km (3 sigma of solution 220)	automatic
C6	Yarkovsky A2 moves > 3 sigma, a radial or normal push becomes significant, or JPL and ESA disagree on A2	automatic
C7	Narrowband radio signal tracking Apophis, confirmed by a second station (2029 flyby)	manual, not yet possible
C8	Light curve, spin or colour inconsistent with a natural body, or close-up imaging	manual, not yet possible

Hypotheses: H0, the agencies' orbit models are correct; H-A (stated by the investigators), Apophis is artificial or controlled. A recorded bet (section 4d) puts the 2029 distance at 28,000 to 29,000 km (investigator) or 38,008 to 38,015 km (AI assistant, from JPL's 3-sigma window).

2 Data and methods

Sources (all public, no authentication): NASA/JPL CNEOS close-approach API, Sentry API and Small-Body Database (orbit and non-gravitational parameters, full precision); NASA/JPL Horizons (topocentric ephemeris for the Lisbon point, 1-minute steps over ±12 h around closest approach); ESA NEOCC close approaches (.clolin), risk list, observation file (.rwo) and orbit (.ke1). Each snapshot fetches every source live (no cache, no stale copy) and stores the replies, their SHA-256 hashes, a manifest and the extracted values. Replies over 200 kB (ESA risk list and observation file, the Horizons table) are kept as hash plus extracted lines.

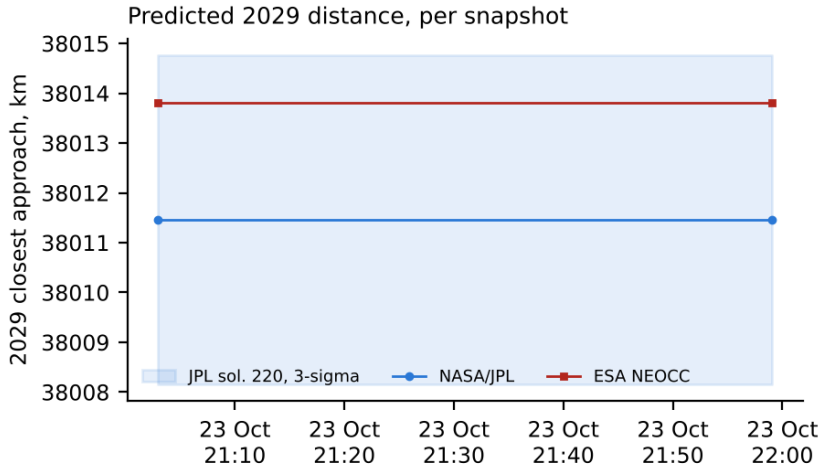
Comparison: passes are matched across agencies (within 5 days, or 3 times ESA's time uncertainty), and JPL's TDB times are converted to UTC (−69.184 s). The combined sigma is √(σ_JPL² + σ_ESA²), with σ_JPL = window/6 from JPL's 3-sigma bounds and σ_ESA ESA's stated 1-sigma. The Lisbon test takes the minimum over the window of the topocentric distance minus Horizons' 3-sigma range uncertainty and compares it with 100 km. Software: `orbital/hannibal_crosscheck.py`, `hannibal_neo_where.py`, `docs/investigation/apophis/collect.py`, all with offline tests.

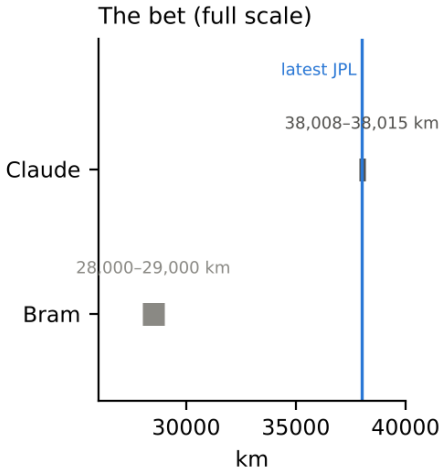
3 Results

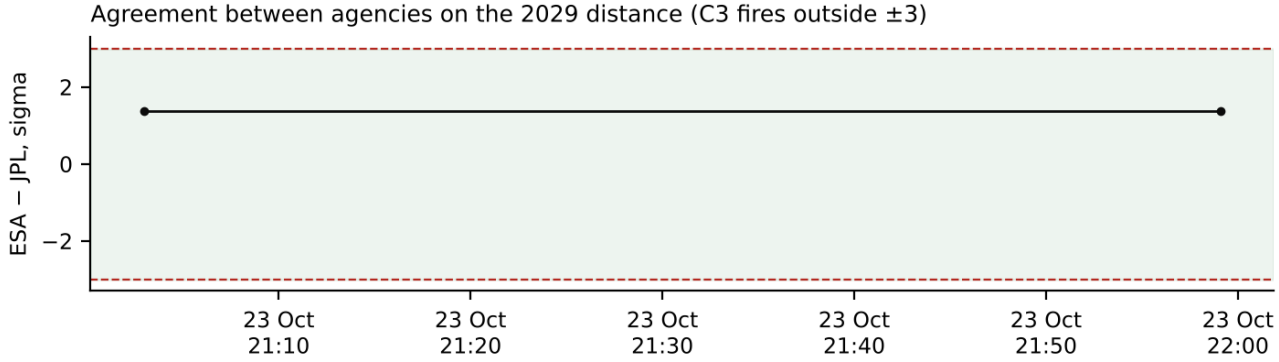
Quantity	n	min	max	mean	stdev
JPL 2029 distance, km	2	38,011.46	38,011.46	38,011.46	0.000
ESA 2029 distance, km	2	38,013.80	38,013.80	38,013.80	0.000
ESA − JPL, km	2	2.35	2.35	2.35	0.000
|ESA − JPL|, sigma	2	1.37	1.37	1.37	0.000
JPL nearest 3-sigma edge, km	2	38,008.18	38,008.18	38,008.18	0.000
ESA minimum possible, km	2	38,009.85	38,009.85	38,009.85	0.000
Time ESA − JPL (UTC), s	2	12.6	12.6	12.6	0.000
Lisbon nearest edge, km	1	32,491.9	32,491.9	32,491.9	0.000

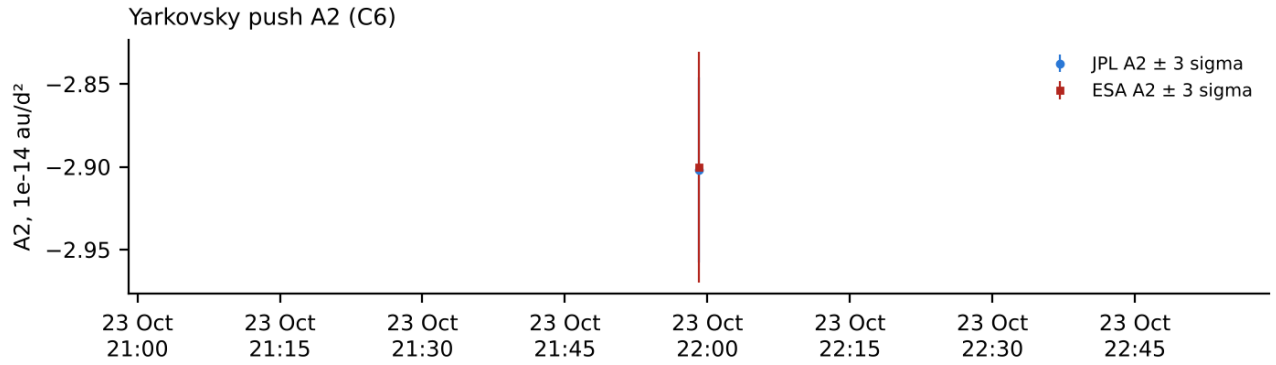
Criterion	fired	checked	fired in snapshots
C1	0	2	–
C2	0	2	–
C3	0	2	–
C4	0	1	–
C5	0	2	–
C6	0	1	–

Orbit solutions seen: 220. Latest observation used by JPL: 2022-04-09; in ESA's file: 2022-04-09. Sentry: removed 2021-02-21 08:22:28. ESA risk list: not listed. Source failures (snapshots): none. Integrity failures: 0.









The bet: the latest JPL value, 38,011.46 km, lies inside Claude's range. The bet is settled only by the post-flyby reconstruction (April 2029).

4 Threats to validity and bias controls

Issue	Effect	Control or statement
Shared observations	JPL and ESA fit largely the same Minor Planet Center observations; their agreement tests the fitting, not the observations themselves.	Stated. The study's own astrometry (hannibal_astrometry_check.py, from Dec 2026) adds independent observations.
Snapshots are not independent	Hourly snapshots of an unchanged orbit solution repeat one measurement. n is the number of looks, not the number of independent samples.	No statistical test treats them as independent. What the series can detect is the publication of a new solution.
Linearised uncertainty	3-sigma windows assume a linear covariance; this fails for distant passes (2116).	C3 ignores passes whose window is larger than the distance; those are logged as notes.
Observation gap	No observations since 2022-04-09; Apophis is unobservable until about 20 Dec 2026.	The null result covers the period only; the study continues through the 2027–2029 windows.
Investigator expectation	The investigators hold H-A and bet against the models.	Criteria, thresholds and conclusion wording were fixed before the data; the pipeline picks the wording.
AI assistance	The software and this report were written with Claude (Anthropic).	All code, data and hashes are public in the repository; anyone can rerun the pipeline.
Data integrity	Files could be altered after collection.	SHA-256 per file, checked on every build; failures are excluded from statistics and listed.

5 Conclusion

No pre-registered criterion fired in any verified snapshot. Over the study period, the public data from NASA/JPL and ESA NEOCC give no evidence for a risk of impact by 99942 Apophis, including at the point off Lisbon, and no evidence for a push the Yarkovsky model doesn't explain. This is a null result for the period covered. It does not test predictions that depend on observations not yet made (Apophis is next observable from about 20 December 2026).

Predictions for the 2029 model test (section 4c of the pre-registration) remain open until the post-flyby reconstruction: closest approach 21:44:50 UTC (JPL) or 21:45:03 UTC (ESA), at 38,011 km (JPL 3-sigma window 38,008 to 38,015 km) or 38,014 km (ESA).

6 Reproducibility

`git clone --branch reports/daily https://github.com/3mm4nu3I/Hannibal-Beta-2.0`
`python docs/investigation/apophis/study/build_study.py --until <same cutoff>`
rebuilds the dataset, figures and this report from the stored evidence. `study/out/STUDY_MANIFEST.json` lists the SHA-256 of every input folder and every output. Each evidence folder can be checked on its own with `sha256sum -c SHA256SUMS`.

7 Data and code availability

All evidence snapshots, the pre-registration, the software and this report's builder are public in the repository above (branch reports/daily, pull request #15). Raw agency replies under 200 kB are stored verbatim; larger ones are stored as SHA-256 hash plus extracted lines and can be re-downloaded from the agencies for comparison. Excluded material: station telemetry summaries dated 20 to 22 October 2026 are not used, because they cannot be reproduced from recorded data.

8 Contributions and AI-use disclosure

Study design, hypotheses H-A and the investigator's prediction: Bram (Project Hannibal). Software, analysis pipeline and drafting of this report: Claude, an AI model by Anthropic, at the investigator's request; the AI assistant's own prediction is recorded separately (pre-registration 4d) and plays no part in the criteria. No data was generated by the AI: every number in the results comes from the agencies' replies stored in the evidence folders.

9 References (data services)

[1] NASA/JPL Center for Near-Earth Object Studies, SBDB Close-Approach Data API, ssd-api.jpl.nasa.gov/cad.api. [2] NASA/JPL Sentry: Earth Impact Monitoring, ssd-api.jpl.nasa.gov/sentry.api. [3] NASA/JPL Small-Body Database API, ssd-api.jpl.nasa.gov/sbdb.api. [4] NASA/JPL Horizons System API, ssd.jpl.nasa.gov/api/horizons.api. [5] ESA NEO Coordination Centre, close approaches, risk list, observations and orbits, neo.ssa.esa.int. [6] SatNOGS Network and DB (Libre Space Foundation), network.satnogs.org and db.satnogs.org (Appendix B).

Appendix A: snapshots

stamp	integrity	orbit	JPL 2029 km	ESA 2029 km	sigma	Lisbon	fired
2026-10-23_2103	OK	220	38,011.46	38,013.80	1.37	–	none
2026-10-23_2159	OK	220	38,011.46	38,013.80	1.37	IMPOSSIBLE	none

Appendix B: supplementary result, dormant military satellites still transmitting

A separate question from the Apophis study, included because it shows the same method (public records, fixed rules, reproducible). On 23 October 2026 the 148 military objects of the investigators' 17 October export were checked against SatNOGS for a vetted good observation (any volunteer station) in the 30 days before. HEARD means at least one such observation; it is not reception by the investigators' own station.

Outcome	Objects
HEARD	2
LISTED, NOT HEARD	3
NO ACTIVE TX	1
NOT CHECKED	0
NOT IN SATNOGS	142

Object	NORAD	Launched	Heard on	Vetted good / bad (30 d)	Last good (UTC)
COSMOS 2488	39483	2013	UHF Unknown, FM, 388.687 MHz	9 / 0	2026-10-21 12:45
ERS 15	2412	1966	carrier wave, CW, 136.439 MHz	2 / 5	2026-09-26 12:40

Source file docs/reports/2026-10-23_mission-report/zombie_23oct.json, SHA-256 6ad4d0b81ea126037b6e439174530a48ebffe5b07d269d0a8c94d2ba7e844dc5. Rerun (with a new 30-day window): `python orbital/hannibal_zombie.py --json out.json`. Limits: 142 of the 148 objects have no SatNOGS record, so for them the check is silent, not negative; ERS 15's 2 good against 5 bad observations is a small sample, consistent with an intermittent, sunlight-dependent transmitter. This result plays no part in the Apophis criteria.